UNIVERSITY OF MALTA - ARI5907 PLACEMENT IN AI

M.AI.ESTRO-TWIN Vision Pipeline

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M.AI.ESTRO - Automated Digital Twin Synthesis for Industrial Skill Transfer

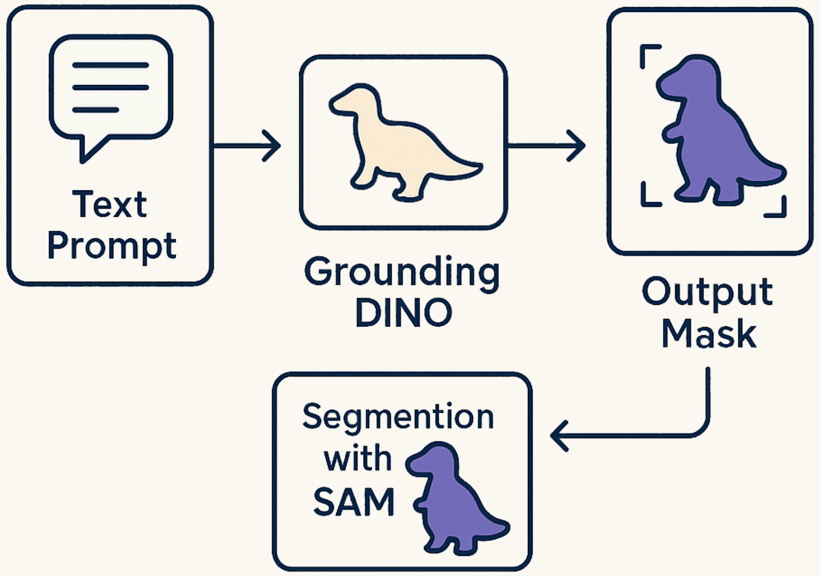
PROBLEM CONTEXT

World Models compress every visible variation

Masks improve scope by filtering out nuisance variations

PROBLEM CONTEXT

Open-set perception requires input prompt, normally manual



RESEARCH QUESTION

Can VLM object lists become automated prompts for open-set grounding?

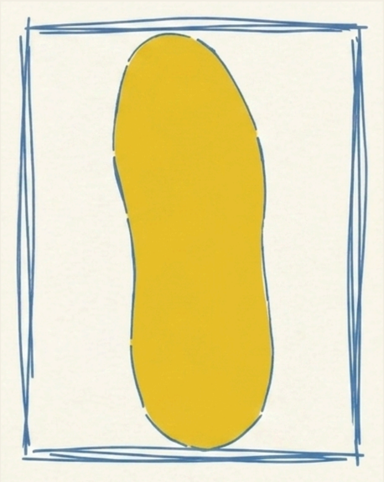
PIPELINE

EGODEX SAMPLE

SEMANTICS

```
caption: shoe handling
objects: shoe, laces
query: shoe
```

BOUNDING BOX & MASK



Each sampled window keeps one aligned record: semantic description, object prompts, boxes, masks, and file manifest.

Real EgoDex frame

VLM Structured Output

caption: hands manipulate a shoe and its laces
object_list: ["shoe", "laces"]
actions: ["grasp", "adjust", "contact"]
contact_object: "shoe"
hand_contact: true

MODEL CHOICE

The selected model had to be good enough and fast enough.

64
shared windows

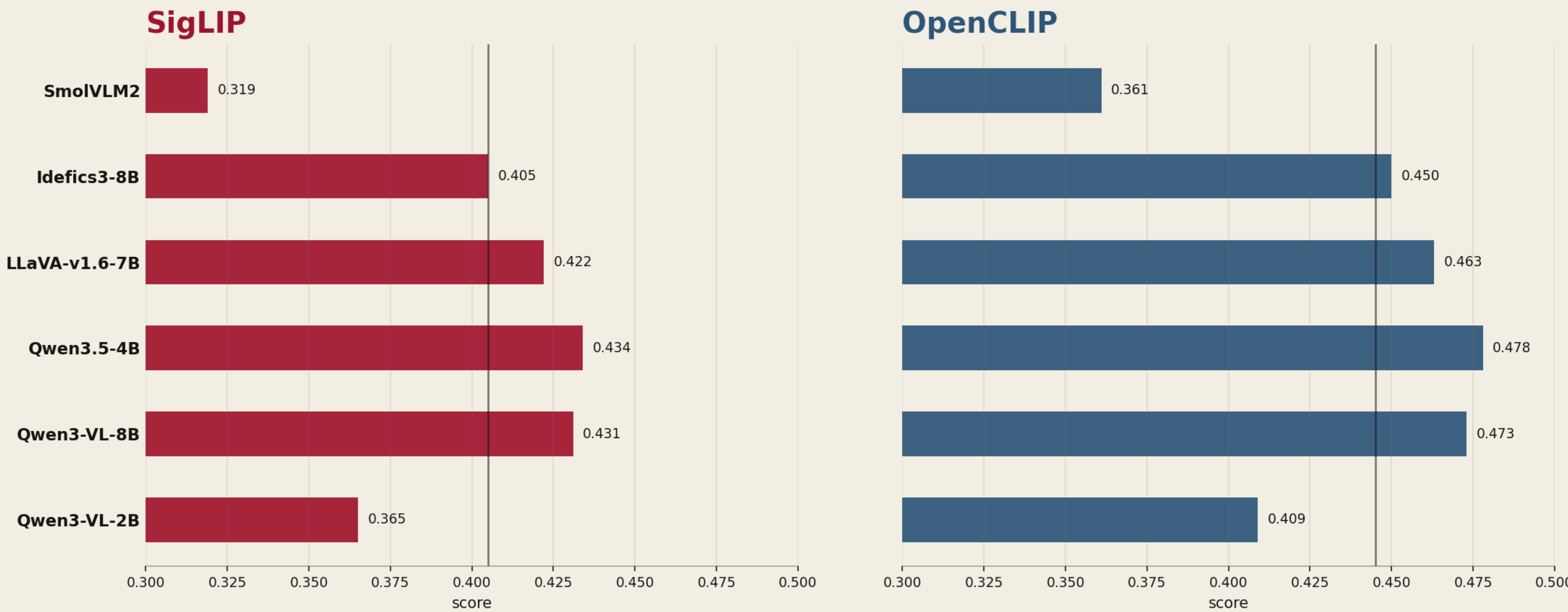
0.365
SigLIP score

1.432
s/window

BENCHMARK RESULT

Qwen3-VL-2B won operationally.

SCORE DISTRIBUTION



PROMPT BRIDGE

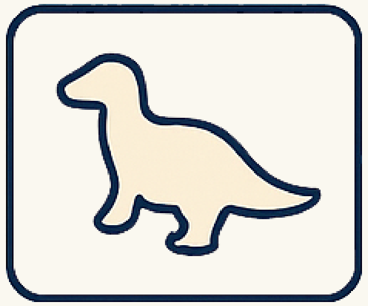
The object list became the handoff.

OBJECT LIST

```
["shoe", "laces"]
```

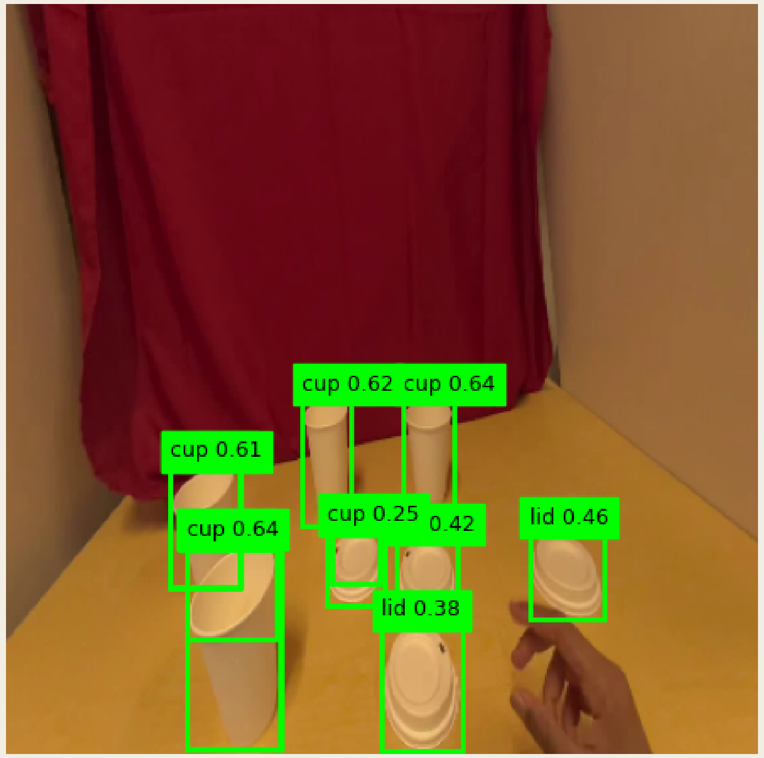
→

GROUNDING DINO



SPATIAL OUTPUTS

BOUNDING BOXES



OBJECT MASKS

ARTIFACT PROOF

SEMANTIC OUTPUTS

captions.jsonl
tags.jsonl
embeddings.npy
manifest.json

GROUNDING AND MASK OUTPUTS

detections.jsonl
masks.jsonl
progress.json
manifest.json

NEXT WORK

Frame perception as world-model enrichment.	**Build and benchmark the enriched variant.**	**Reuse for DPP Malta-Poland Call.**